# B. Tech ECE Working Model 2
# Final Review

## Analog Circuits (BECE206L)

IOT Based Air Quality Monitoring Index


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# LITERATURE SURVEY



**IoT in Environmental Monitoring**

IoT-based monitoring systems are widely used for real-time data collection in environmental applications, providing valuable insights into air quality, pollution levels, and weather conditions.



**BMP208 Sensor for Atmospheric Data**

BMP208 sensors are reliable for measuring temperature and atmospheric pressure, widely adopted in weather monitoring and environmental research due to high accuracy and low power consumption.



**MQ2 Sensor for Gas Detection**

The MQ2 sensor is commonly used for detecting gas concentrations like LPG, smoke, and methane, making it ideal for air quality systems. Previous studies validate its effectiveness in monitoring indoor and outdoor pollution.



**Blynk IoT Platform for Data Visualization**

Blynk has proven effective for real-time monitoring in IoT projects, allowing easy access to data from remote locations. It is commonly used in similar air quality monitoring projects for displaying metrics on mobile devices.



**Data Logging for Air Quality**

Continuous data logging, as implemented with SD cards, supports analysis over time, enabling trend identification and response planning for air quality management.

# IDENTIFYING THE GAP

| | |
|---|---|
| **Limited Real-Time Monitoring with Data Storage** | • Many air quality monitoring systems provide real-time data but lack continuous data logging for historical analysis. Our system addresses this by storing readings every 5 seconds on an SD card. |
| **Few Cost-Effective, Portable Solutions** | • Existing systems can be costly and complex, making them less accessible. Our solution is affordable, portable, and easy to implement using ESP32, MQ2, and BMP208 sensors. |
| **Absence of Comprehensive Gas Monitoring in Small Devices** | • Most compact devices focus on limited air quality metrics. By integrating multiple metrics (temperature, pressure, LPG, gas, and smoke levels), our system provides a fuller picture of air quality in a single, small device. |
| **Need for User-Friendly IoT Integration** | • While many air quality monitors exist, few offer seamless IoT integration for remote monitoring. Our system uses Blynk, enabling users to monitor air quality on mobile devices conveniently. |

# PROBLEM STATEMENT

*Air pollution and poor indoor air quality pose significant health risks, yet accessible, affordable, and comprehensive monitoring solutions are limited. Current systems often lack real-time data accessibility, reliable storage for historical data analysis, or user-friendly remote monitoring. This project aims to develop an IoT-based air quality monitoring system that provides real-time insights into temperature, pressure, LPG levels, gas concentration, and smoke levels, with continuous data logging and remote monitoring through a mobile app.*

# PROJECT OBJECTIVE

To design and implement an IoT-based air quality monitoring system that:

- **Measures Key Air Quality Metrics**
  - Monitors temperature, pressure, LPG level, gas concentration, and smoke level using BMP208 and MQ2 sensors.
- **Provides Real-Time Data Access**
  - Utilizes Blynk for displaying live data on a mobile app for remote monitoring.
- **Enables Data Logging for Analysis**
  - Records all sensor readings on an SD card every 5 seconds, allowing for historical data analysis and trend tracking.
- **Offers a Cost-Effective and Portable Solution**
  - Uses affordable components to create a compact and accessible device suitable for various settings.



# DESIGN APPROACH AND ALGORITHM

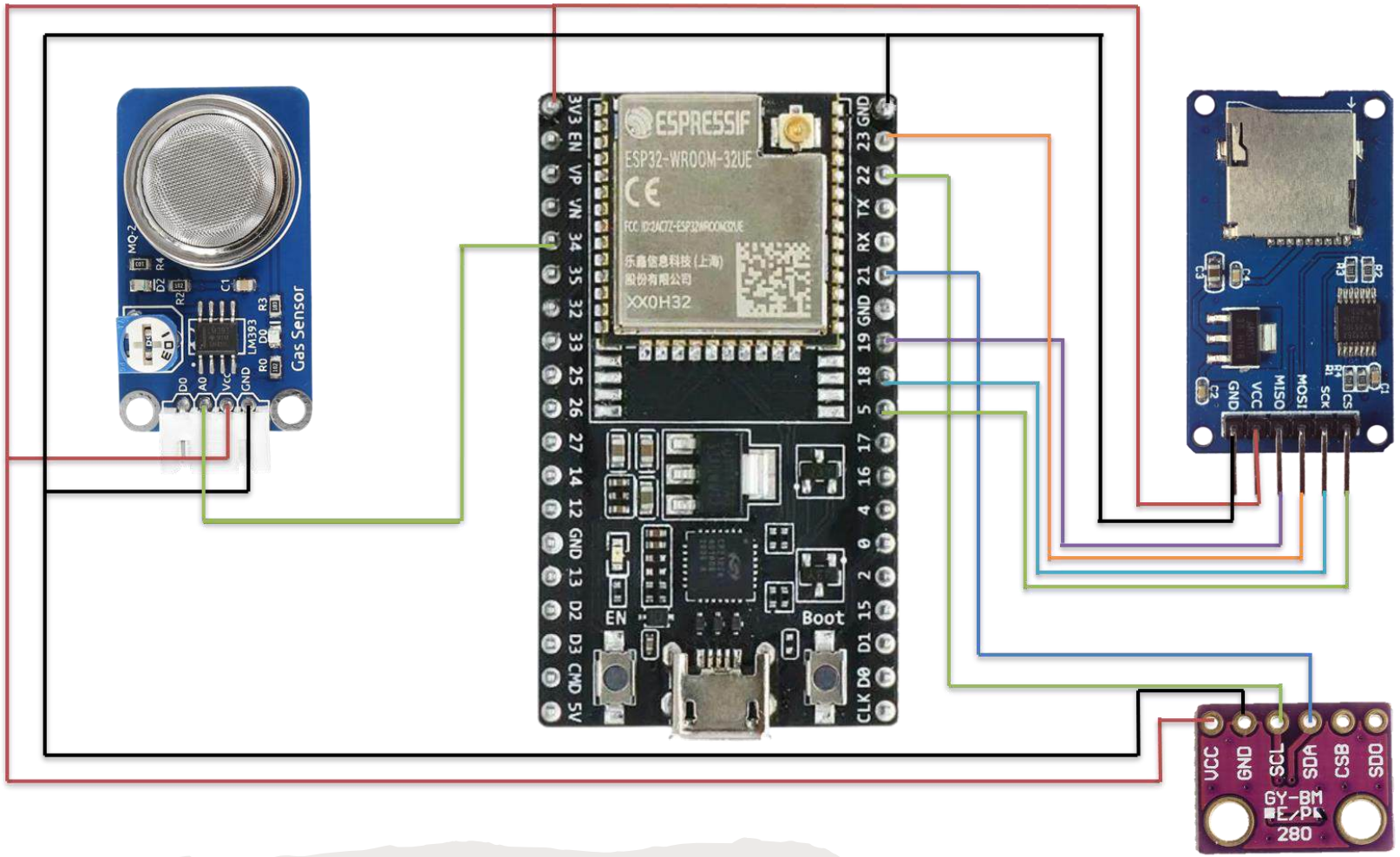
## Design Approach

- **ESP32**: Main controller, handles data from sensors, sends data to Blynk, logs data on SD card.
- **BMP208 Sensor**: Measures temperature and pressure.
- **MQ2 Sensor**: Measures LPG, gas concentration, and smoke.
- **SD Card Module**: Stores data every 5 seconds for record-keeping and analysis.

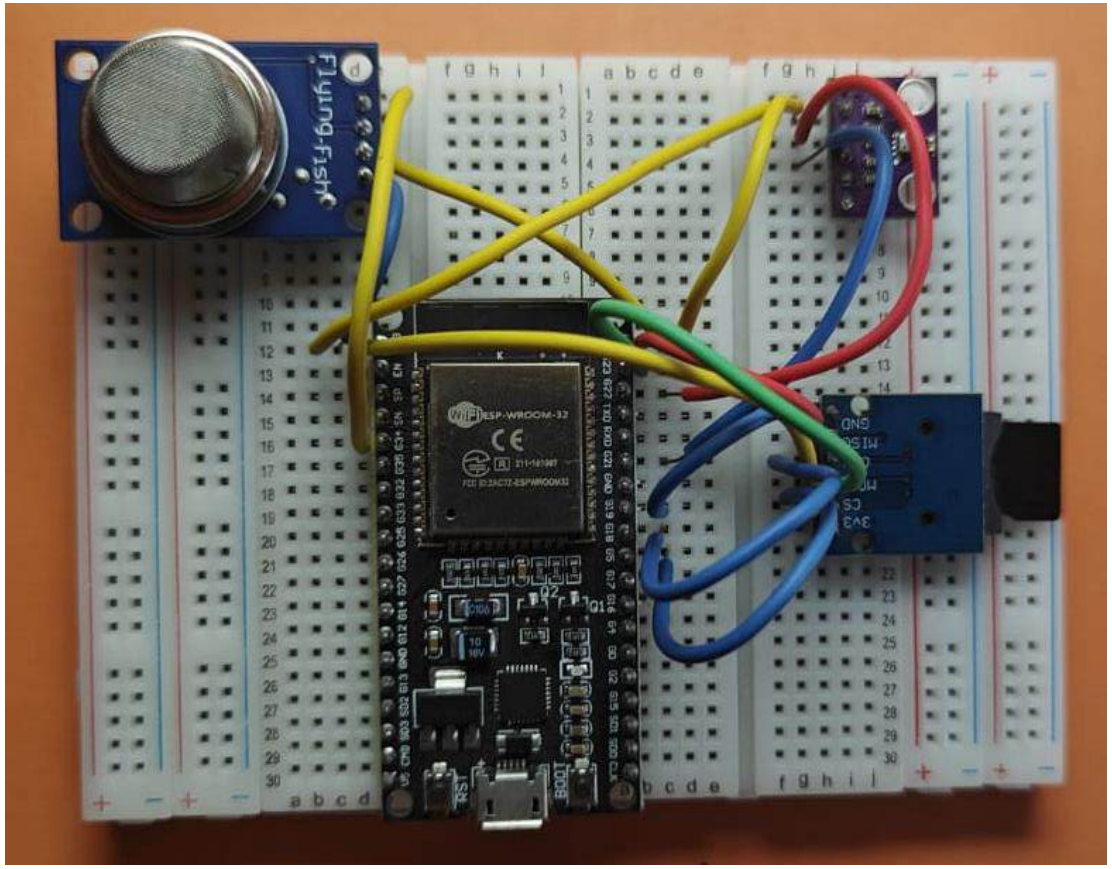
## Algorithm

- **Step 1**: Initialize sensors and SD card.
- **Step 2**: Read data from BMP208 and MQ2 sensors.
- **Step 3**: Display data on Blynk app in real time.
- **Step 4**: Save data to SD card every 5 seconds.
- **Step 5**: Repeat.

# SYSTEM ARCHITECTURE




# Project Circuit




# Project Illustration

ESP32_AQMI.ino

```
#define BLYNK_TEMPLATE_ID "TMPL3G4CizG4Q"
#define BLYNK_TEMPLATE_NAME "AIR QUALITY MONITORING INDEX"
#define BLYNK_AUTH_TOKEN "63SQ-G0uny2spuYHcAp14YDRY2_fb9qI"

#define BLYNK_PRINT Serial
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

#include <Adafruit_BMP280.h>
#include <SPI.h>
#include <SD.h>

// Initialize Blynk and WiFi credentials
char auth[] = BLYNK_AUTH_TOKEN;
char ssid[] = "OnePlus 11 5G";  // Your WiFi name
char pass[] = "harikruthik";    // Your WiFi password

// Pin Definitions
const int gasPin = 34;  // MQ-02 sensor connected to GPIO34 (ADC)
const int csPin = 5;    // SD card module CS pin

// BMP280 I2C address and sensor setup
Adafruit_BMP280 bmp;
```

ESP32_AQMI.ino

```

// SD card setup
File dataFile;

// Function to initialize BMP280 sensor
bool initBMP280() {
    if (!bmp.begin(0x76)) {
        Serial.println("Could not find a valid BMP280 sensor, check wiring!");
        return false;
    }
    return true;
}

// Function to read MQ-02 sensor value (Analog) and calculate gas concentrations
int readGasConcentration() {
    return analogRead(gasPin);
}

int readSmokeLevel() {
    return analogRead(gasPin);
}

int readLPGLevel() {
    return analogRead(gasPin);
}
```

# Project Codes

ESP32_AQMI.ino

```
}

// Function to log data to the SD card
void logDataToSD(String data) {
    dataFile = SD.open("/air_quality_data.txt", FILE_WRITE);
    if (dataFile) {
        dataFile.println(data);
        dataFile.close();
        Serial.println("Data logged to SD card.");
    } else {
        Serial.println("Error opening SD card file.");
    }
}

// Send sensor data to Blynk, log it, and display it on the serial monitor
void sendSensorData() {
    float temperature = bmp.readTemperature();
    float pressure = bmp.readPressure() / 100.0F;
    int gasConcentration = readGasConcentration();
    int smokeLevel = readSmokeLevel();
    int lpgLevel = readLPGLevel();

    // Send data to Blynk virtual pins
    Blynk.virtualWrite(V0, temperature);         // Temperature
```

ESP32_AQMI.ino

```
    Blynk.virtualWrite(V0, temperature);         // Temperature
    Blynk.virtualWrite(V1, pressure);            // Pressure
    Blynk.virtualWrite(V2, gasConcentration);    // Gas concentration
    Blynk.virtualWrite(V3, smokeLevel);          // Smoke level
    Blynk.virtualWrite(V4, lpgLevel);            // LPG level

    // Display data in the serial monitor
    Serial.print("Temperature (C): ");
    Serial.println(temperature);
    Serial.print("Pressure (hPa): ");
    Serial.println(pressure);
    Serial.print("Gas Concentration: ");
    Serial.println(gasConcentration);
    Serial.print("Smoke Level: ");
    Serial.println(smokeLevel);
    Serial.print("LPG Level: ");
    Serial.println(lpgLevel);
    Serial.println();

    // Prepare data string for logging
    String data = "Temperature: " + String(temperature) +
                  ", Pressure: " + String(pressure) +
                  ", Gas Concentration: " + String(gasConcentration) +
                  ", Smoke Level: " + String(smokeLevel) +
```

# Project Codes

ESP32_AQMI.ino

```
                ", Smoke Level: " + String(smokeLevel) +
                ", LPG Level: " + String(lpgLevel);

    // Log data to SD card
    logDataToSD(data);
}

void setup() {
    Serial.begin(115200);
    Serial.println("Starting setup...");

    // Initialize WiFi
    Serial.print("Connecting to WiFi: ");
    Serial.println(ssid);
    WiFi.begin(ssid, pass);

    int attempts = 0;  // To count attempts
    while (WiFi.status() != WL_CONNECTED && attempts < 20) {  // Limit attempts to avoid infinite loop
        delay(500);
        Serial.print(".");
        attempts++;
    }

    if (WiFi.status() == WL_CONNECTED) {
        Serial.println("\nConnected to WiFi");
```

ESP32_AQMI.ino

```
    }

    if (WiFi.status() == WL_CONNECTED) {
        Serial.println("\nConnected to WiFi");
        Serial.print("IP Address: ");
        Serial.println(WiFi.localIP());
    } else {
        Serial.println("\nFailed to connect to WiFi. Check SSID and password.");
        Serial.print("WiFi status: ");
        Serial.println(WiFi.status()); // Print the current WiFi status
    }

    // Initialize Blynk
    Blynk.begin(auth, ssid, pass);
    Serial.println("Blynk initialized");

    // Initialize BMP280 sensor
    if (!initBMP280()) {
        Serial.println("BMP280 initialization failed!");
    } else {
        Serial.println("BMP280 initialized successfully.");
    }

    // Initialize SD card
    if (!SD.begin(csPin)) {
```

# Project Codes

ESP32_AQMI.ino

```
    Blynk.begin(auth, ssid, pass);
    Serial.println("Blynk initialized");

    // Initialize BMP280 sensor
    if (!initBMP280()) {
        Serial.println("BMP280 initialization failed!");
    } else {
        Serial.println("BMP280 initialized successfully.");
    }

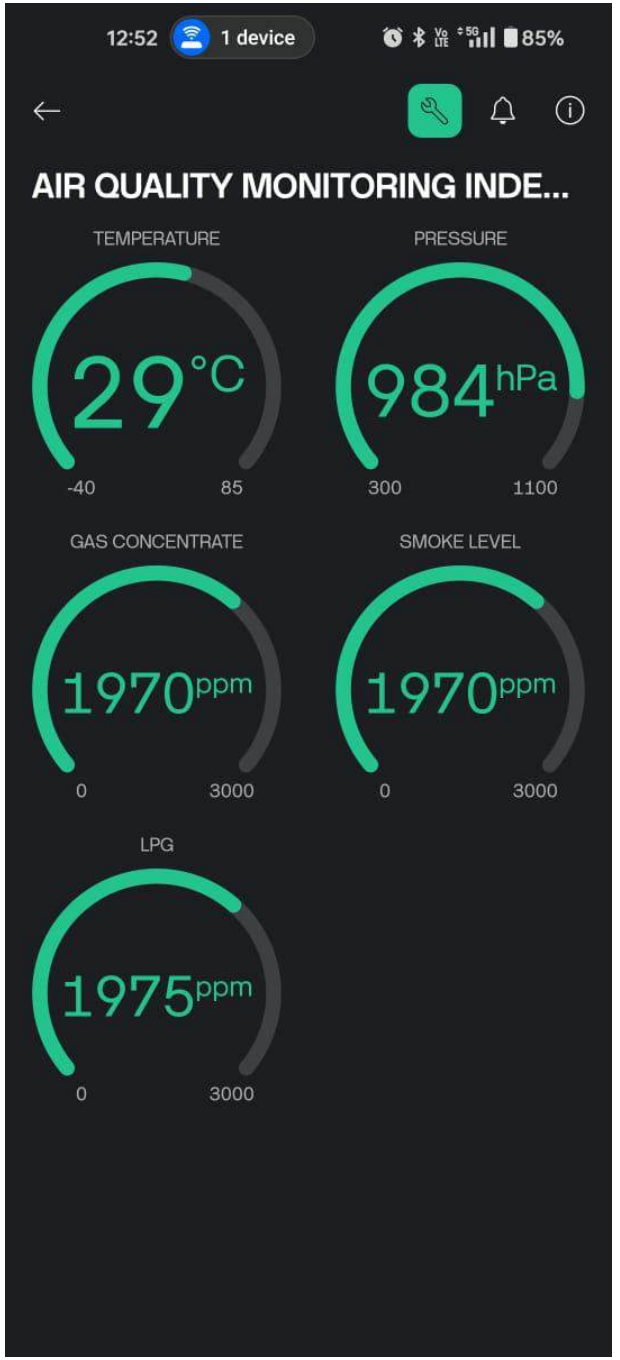
    // Initialize SD card
    if (!SD.begin(csPin)) {
        Serial.println("SD card initialization failed!");
    } else {
        Serial.println("SD card initialized successfully.");
        logDataToSD("Temperature (C), Pressure (hPa), Gas Concentration, Smoke Level, LPG Level");
    }
}

void loop() {
    Blynk.run();
    sendSensorData();  // Read and send sensor data to Blynk
    delay(5000);       // Delay for 5 seconds between readings
}

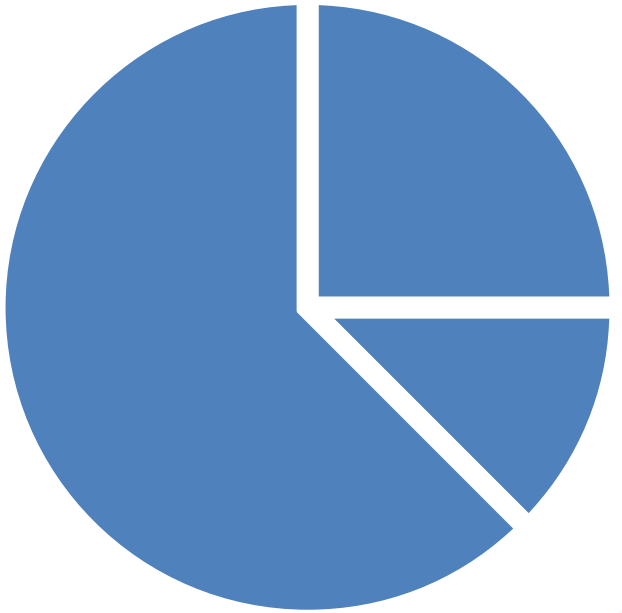
```

# Project Codes

# BLYNK UI


12:52
1 device
85%
AIR QUALITY MONITORING INDE...
TEMPERATURE
29°C
-40
85
PRESSURE
984hPa
300
1100
GAS CONCENTRATE
1970ppm
0
3000
SMOKE LEVEL
1970ppm
0
3000
LPG
1975ppm
0
3000

# ANALYTICAL AND THEORETICAL DESCRIPTION



1. **Sensor Data Interpretation**

   **BMP208 Sensor**: Uses barometric pressure and temperature sensors based on piezo-resistive technology. Outputs digital values that correspond to environmental temperature and atmospheric pressure levels.

   **MQ2 Sensor**: The MQ2 sensor operates based on a change in resistance when exposed to gases like LPG and smoke. The sensor outputs a voltage proportional to the concentration of these gases, allowing for an estimation of air pollution levels.

2. **Data Processing and Calibration**

   **Temperature and Pressure Calculation**: The BMP208's output values are converted to Celsius and Pascal units based on factory calibration.

   **Gas Concentration Estimation**: The MQ2 sensor provides an analog value, which is converted to gas concentration (ppm) using a specific sensitivity curve and calibration factor.

3. **Data Logging Mechanism**

   **Sampling Frequency**: Data is sampled every 5 seconds and stored on the SD card. This enables time-stamped historical analysis.

   **Data Transmission to Blynk**: The ESP32 sends processed sensor data to Blynk via Wi-Fi, allowing real-time monitoring.

4. **Theoretical Basis for System Design**

   **IoT in Air Quality Monitoring**: Theoretical studies show that IoT systems enhance real-time environmental monitoring. Blynk is leveraged to streamline data visualization, following principles of remote sensing and IoT-based environmental analysis.

# HARDWARE/SOFTWARE TOOLS USED

- **ESP32** – Microcontroller for processing sensor data and connecting to Blynk via Wi-Fi.
- **BMP208 Sensor** – Measures temperature and atmospheric pressure.
- **MQ2 Sensor** – Detects LPG, gas concentration, and smoke levels.
- **SD Card Module** – Stores data every 5 seconds for record-keeping.
- **Small Breadboard –** Keeps all the components intact and helps connect the components
- **Power Bank** – Provides temporary power supply via ESP32's micro-USB socket for demonstrations.
- **Software Tools**
- **Arduino IDE** – Used for coding and uploading firmware to ESP32.
- **Blynk App** – IoT platform to monitor real-time data on mobile devices.

# RESULT ANALYSIS

- **Real-Time Monitoring**
  - Successful live monitoring of temperature, pressure, LPG level, gas concentration, and smoke level on the Blynk app, demonstrating effective IoT integration.
- **Data Logging**
  - Data recorded on the SD card every 5 seconds, providing a continuous log for historical analysis.
- **System Responsiveness**
  - System showed prompt response to changes in air quality, with accurate readings displayed in real-time on the mobile app.
- **Testing and Validation**
  - A video of the project test has been recorded, demonstrating system functionality and response.

*Ctrl + Click here to redirect to the Video*

*(or copy paste link):*
*https://drive.google.com/file/d/1q7RRBdI7jXQhRD2iEzz6ik8fPcAbJSuv/view?usp=sharing*

# CONCLUSION AND FUTURE WORK

## Conclusion

- Successfully developed an IoT-based air quality monitoring system capable of tracking temperature, pressure, LPG, gas concentration, and smoke levels in real-time with data logging for analysis.

## Limitations and Constraints

- Limited sensor range due to cost efficiency may affect accuracy in larger or open spaces.
- Requires continuous Wi-Fi connection for real-time monitoring on Blynk.

## Future Improvements

- **Additional Sensors** – Add sensors for more pollutants, such as CO2 and particulate matter.
- **Data Analysis and Alerts** – Implement data analysis features and real-time alert notifications for critical air quality levels.

# Social and Environmental Impact



**Health and Safety Awareness**

Promotes public awareness of indoor air quality, helping users identify harmful levels of pollutants that can affect respiratory health, especially for vulnerable populations.



**Environmental Monitoring**

Supports efforts to monitor and reduce air pollution by providing accessible, real-time data on key environmental metrics.



**Use Case During Diwali**

During Diwali, increased fireworks lead to a spike in air pollution levels. This system allows households to monitor air quality in real-time, helping them make decisions about ventilation, mask usage, or limiting outdoor exposure, especially for children, the elderly, and individuals with respiratory issues.



**Contribution to Sustainable Living**

Enables individuals and communities to take informed actions to improve air quality, contributing to a healthier and more sustainable environment.



**Cost-Effective Solution**

Provides an affordable way to track air quality, making it accessible to households, schools, and small businesses focused on creating safer environments.

# WORK PLAN

| PHASE | TIME PERIOD | TASKS | CONTRIBUTION |
|---|---|---|---|
| Phase 1 | Before CAT1 Exams | Planning and discussing the project ideas. | Collaborative brainstorming and idea generation by all team members. |
| Phase 2 | After CATI and before CATII Exams | Finalized the project topic, researched sensors, coding requirements, and purchased components. | Each member researched specific areas (sensors, coding, hardware) and pooled findings. |
| Phase 3 | After CATII | Assembled all components, resolved bugs, and completed the project to perfection. | Worked together on assembly, debugging, and finalization for a successful implementation. |



# INDIVIDUAL CONTRIBUTION

**23BEC0124 – SRIGANTH C**

- **Project Planning**: Contributed to brainstorming and discussing project ideas and objectives during Phase 1.
- **Research on Sensors**: Took the lead in researching suitable sensors (BMP208 and MQ2) for the air quality monitoring system.
- **Component Purchase**: Responsible for sourcing and purchasing necessary components for the project.

**23BEC0358 – HARI KRUTHIK**

- **IoT Integration**: Took charge of integrating the IoT platform (Blynk) and ensuring real-time data transmission to the mobile app.
- **Debugging and Bug Fixing**: Worked on resolving bugs related to the hardware and software integration during Phase 3.
- **Final Presentation**: Assisted in preparing the final project report and presentation, ensuring clarity and coherence.

**23BEC0212 – DHAKSHANA BALA S**

- **System Design**: Contributed to system architecture and design, ensuring proper integration of ESP32, sensors, and SD card module.
- **Coding**: Developed the code to interface with the BMP208 and MQ2 sensors, as well as for data logging and communication with Blynk.
- **Testing**: Performed initial system tests and identified issues related to sensor calibration and data accuracy.

# COST ANALYSIS

| Component | Description | Cost (INR) |
|---|---|---|
| **ESP32** | Microcontroller for processing and Wi-Fi connectivity. | 300 |
| **BMP208 Sensor** | Sensor for measuring temperature and atmospheric pressure. | 200 |
| **MQ2 Sensor** | Sensor for detecting LPG, gas concentration, and smoke. | 150 |
| **SD Card Module** | For storing data logs every 5 seconds. | 100 |
| **Power Bank** | Temporary power supply for testing. | 500 |
| **Wires and Connectors** | For connecting components and assembly. | 50 |
| **Miscellaneous** | Small additional items (PCB, casing, etc.). | 200 |

**Total Estimated Cost**: ₹1,500 (approx.)

# REFERENCES


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3. J. R. Smith, K. T. Johnson, and L. L. Davis, "MQ Series Sensors: Understanding Gas Detection Technology," *IEEE Transactions on Environmental Technology*, vol. 12, no. 1, pp. 100-110, Jan. 2018, doi: 10.1109/ET.2018.8765345.
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